

Ronan C. Corgel

INFECTIOUS DISEASE EPIDEMIOLOGY DOCTORAL RESEARCHER

📍 Washington, DC | ✉️ rcc92@georgetown.edu | 🌐 rcorgel.com | 📷 rcorgel | in rcorgel

Summary

I'm currently a fourth-year PhD candidate in Georgetown University's Global Infectious Disease program where I'm advised by Dr. Shweta Bansal and am a member of the Bansal Lab. My research interests broadly include the mathematical modeling of infectious disease dynamics, public health surveillance, and the impact of human behavior on disease transmission. After completing my doctorate, I hope to leverage my epidemiology and data science training to work in an applied public health setting.

Education

Georgetown University

Washington, DC

DOCTOR OF PHILOSOPHY (PHD) IN GLOBAL INFECTIOUS DISEASE

Aug. 2023 - Exp. May 2028

- **Advisor:** Dr. Shweta Bansal
- **Dissertation Title:** *"Toward robust respiratory virus preparedness: big data, behavior, and surveillance"*

Summer Institute in Statistics and Modeling in Infectious Diseases

Atlanta, GA

SUMMER INSTITUTE STUDENT

Jul. 2024

- **Coursework:** MCMC for Infectious Diseases Modeling I, MCMC for Infectious Diseases Modeling II, Simulation-Based Inference for Epidemiological Dynamics

Johns Hopkins Bloomberg School of Public Health

Baltimore, MD

MASTER OF SCIENCE (SCM) IN EPIDEMIOLOGY

Aug. 2021 - May 2023

- **Advisor:** Dr. Amy Wesolowski
- **Thesis Title:** *"Characterizing the Impact of Integrating Spatially Aggregated Human Mobility Data into Infectious Disease Models"*

Johns Hopkins University

Baltimore, MD

BACHELOR OF ARTS (BA) IN PUBLIC HEALTH STUDIES & ECONOMICS

Aug. 2014 - May 2018

- Senior focus in Biostatistics at the Johns Hopkins Bloomberg School of Public Health
- **Activities:** Four-year member of the Johns Hopkins Men's Varsity Swimming Team, JHU Consulting Club

Professional Experience

Centers for Disease Control and Prevention

Baltimore, MD

ORISE FELLOW

Jul. 2023 - Aug. 2023

- Joined the CDC as a summer ORISE Fellow at the newly created Center for Forecasting and Outbreak Analytics (CFA)
- Primarily worked on projects involving public health surveillance of COVID-19 and influenza hospitalizations

U.S. News & World Report

Washington, DC

HEALTH DATA ANALYST

Sept. 2019 - Jul. 2021

- Utilized data science and statistical techniques to evaluate the quality of care provided by hospitals and nursing homes across the United States
- Derived patient care outcomes, created public-facing dashboards, and co-authored data journalism articles

Charles River Associates

Washington, DC

ANTITRUST & COMPETITION ECONOMICS ANALYST

Jul. 2018 - Sept. 2019

- Performed data analyses for companies and law firms regarding mergers, acquisitions, and antitrust litigation disputes
- Main cases focused on anti-competitive drug pricing and monopolization of health care data

Research Experience

Georgetown University

GRADUATE RESEARCHER

Washington, DC

Aug. 2023 - Present

- Conducting research on how social behavior and spatial dynamics shape infectious disease transmission under the guidance of Dr. Shweta Bansal
- Working on projects involving the syndromic surveillance of respiratory viruses, resident-visitor mixing patterns that facilitate disease spread, and forecasting surges in disease activity

Johns Hopkins Bloomberg School of Public Health

GRADUATE RESEARCHER

Baltimore, MD

Oct. 2021 - May 2023

- Worked with Dr. Amy Wesolowski in the Infectious Disease Dynamics research group to analyze data on human mobility and develop infectious disease models
- Examined asymmetric travel patterns in Sub-Saharan Africa, modeled disease dynamics from measles vaccination in Madagascar, and explored mobility patterns relevant to dengue spread in Sri Lanka

Publications

PREPRINT

1. **Corgel, R.**, Tiu, A., Bansal, S. (2025). "Demonstrating the utility of influenza syndromic surveillance with high-volume electronic medical claims in the United States" *medRxiv*. <https://doi.org/10.64898/2025.12.03.25341530>

PUBLISHED

1. **Corgel, R.**, Grantz, K. H., Gardner, L. et al. (2026). "Characterizing the impact of incorporating spatially aggregated human mobility data into infectious disease models." *Proceedings of the National Academy of Sciences*. <https://www.pnas.org/doi/10.1073/pnas.2533465123>
2. Kostandova, N., **Corgel, R.**, Bansal, S. et al. (2025). "Improving mobility data for infectious disease research." *Nature Human Behavior*. <https://doi.org/10.1038/s41562-025-02151-3>

Presentations

2026 **Distinguishing influenza from co-circulating pathogens for improved syndromic surveillance.**

Options XIII: Conference for the Control of Influenza, *Washington, DC*, Poster.

Beyond mobility: who mixes with whom in metapopulations.

The Mathematics of Behavior: an Interdisciplinary School, *Nottingham, UK*, Contributed Talk.

Shifting the scales: how mobility aggregation and social mixing patterns shape epidemic dynamics.

Sorbonne Université Modeling Outbreaks Center, *Paris, FR*, Invited Talk.

2025 **Demonstrating the utility of influenza syndromic surveillance with high-volume medical claims.**

Epidemics 10: International Conference on Infectious Disease Dynamics, *San Diego, CA*, Poster.

Characterizing resident-visitor mixing for spatial disease transmission dynamics.

Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting, *Bethesda, MD*, Scientific Presentation.

2024 **Syndromic surveillance of influenza using medical claims data.**

Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting, *Silver Spring, MD*, Speed Talk & Poster.

Characterizing the impact of incorporating spatially aggregated human mobility data into infectious disease models.

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID), *Atlanta, GA*, Poster.

Teaching

Summer Institute in Statistics and Modeling in Infectious Diseases

Virtual

TEACHING ASSISTANT

Jul. 2025, Jun. 2026

- Assisted in the online module Spatial and Metapopulation Modeling, co-taught by Dr. Shweta Bansal and Dr. Vittoria Colizza
- Developed laboratory assignments, led laboratory sessions, and provided general course support

Johns Hopkins Bloomberg School of Public Health

Baltimore, MD

TEACHING ASSISTANT

Jan. 2023 - Mar. 2023

- Served as a teaching assistant for Concepts and Methods in Infectious Disease Epidemiology, a core infectious disease epidemiology class for master's and Ph.D. students, taught by Dr. David Dowdy
- Provided classroom support, held weekly office hours, and graded student homework assignments

Professional Activities

ORGANIZATION MEMBERSHIP

2023-26 **Member**, Models of Infectious Disease Agent Study (MIDAS) Network

United States

MENTORSHIP

2024-26 **Mentor**, Managed and advised the research project of an undergraduate research assistant

Washington, DC

2022-23 **Peer Mentor**, Provided advice and guidance to first-year master's students

Baltimore, MD

ACADEMIC SERVICE

2022-23 **Master's Student Representative**, Johns Hopkins University Department of Epidemiology

Baltimore, MD

2021-23 **Member**, Epidemiology Student Organization (ESO)

Baltimore, MD

Honors & Awards

2024 **Halloran Travel Scholarship**, Summer Institute in Statistics and Modeling in Infectious Diseases

Atlanta, GA

2023 **Service Recognition Award**, Johns Hopkins University Department of Epidemiology

Baltimore, MD

2018 **General Honors**, Johns Hopkins University

Baltimore, MD

2016-18 **Dean's List**, Johns Hopkins University

Baltimore, MD

2016-18 **National College Athlete Honor Society**, Chi Alpha Sigma

Baltimore, MD

Technical Skills

Programming R (Proficient), Stata (Proficient), Python (Intermediate), SQL (Intermediate)

Geospatial Software ArcGIS (Intermediate), QGIS (Beginner)

Other Git, Microsoft Office Suite (Excel, PowerPoint, Word), Tableau